DRAINCHECK

How DrainCheck works

1 m inspection · 5 m analysis

Real terrain, flood evidence and editable storm scenarios.

Source data

01 REAL TERRAIN
Municipal 1 m DTM
Baltic 1982 ground elevations.

02 OBSERVED / MAPPED
Past floods + buildings
136 flood polygons + cadastral footprints.

03 USER-SET SCENARIO
Area + existing pipes
Draw area; choose pipes and storm.
Drainage and blockage are assumed.

Processing

ALIGNED 5 m GRID
Terrain factors
Elevation · fill depth · slope · flow area

PERCENTILE RANK
Ponding signal P
Rank positive fill depth; zero if none.

P uses fill depth only; slope and upstream contributing area remain diagnostics.

P

SEPARATE REVIEW RULES
Combine P, flood evidence and building overlap
Historical categories override lower terrain classes; buildings flag potential overlap.

History changes review categories; it does not change the terrain heatmap.

ASSUMED WATER BALANCE
Excess runoff
Baseline: max(0, rain×c − rate×min/60)
Clogged: rate×(1−b/100) in pipe buffer.
Baseline removal applies elsewhere.

USES REAL 5 m TERRAIN
Local depression storage
Real bed + fill depth; cap at spill level.
Store each depression's local runoff.

Overflow is not routed; runoff on zero-fill land remains unallocated.

Inputs: rain (mm) · duration (min) · runoff coefficient c · assumed drainage rate (mm/h) · blockage b (%) · pipe buffer (m)

Layers

TERRAIN LAYER
Continuous heatmap
P: 0–100 · not flood probability.

EVIDENCE LAYERS
Review priorities
Independent historical-flood overlays.

STORM LAYERS
Baseline / clogged / change
Est. depth (m) · excess runoff (mm)
Stored · overflow · unallocated (m^3)

DC
MAP
DrainCheck

LAND RULES

Exclude mapped water and uncertain terrain.
Assume mapped greens absorb all rain.

RIVER CONTEXT

River distance: adjacent / away context.
Proximity does not establish flood cause.

SUPPLEMENTAL LAYERS

Archived daily rain, assessments and infrastructure.
Archive rain does not set the storm slider.

Comparative screening and local storage estimates; no dynamic routing, sewer hydraulics or river-level simulation.